Replication of experiment 4b from “Multidimensional visual statistical learning” by Turk-Browne & Isola & Scholl & Treat (2008, *JEP:LMC*)

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Introduction

Visual statistical learning (VSL) is a form of implicit learning. Visual stimuli are repeatedly presented to a participant with certain statistical regularities in the stimulus sequence, for example, stimulus B always occurring after stimulus A. Participants later report such sequences to be more familiar without necessarily being consciously aware of the statistical structure. Turk-Browne and colleagues explored VSL with multidimensional objects. Specifically, they were interested in whether VSL can only occur with objects as a whole, or if it can also operate over different feature dimensions of an object independently. To that end, they report five experiments that used simple abstract objects with two feature dimensions, shape and color. For example, when the red square always follows the blue triangle, does the participant also learn that red follows blue and square follows triangle?

Each experiment consisted of a learning phase where the whole repetition sequence is shown to the participant, and a test phase where short parts of the whole sequence are pitted against “foil” sequences that never occurred in that order during the learning phase. Participants then had to decide which sequence was more familiar to them. The five reported experiments systematically vary the correlation between shape and color (are red objects always squares and squares always red?), or remove either the shape or the color dimension at test. Experiment 4b, which was chosen for replication, had a reduced correlation of shape and color during the learning phase and removed the shape dimension during the test phase (that is, sequences of color patches were presented that did not have a previously shown shape). The VSL effect still occurred, but was weaker than in experiment 1.

Methods

Power Analysis

Turk-Browne et al. (2008) observed a large effect for their primary finding, $t(7) = 2.892$, $p = .023$, $d = 1.023$ (p. 404). At this effect size, 10, 13, 15 participants are needed to achieve 80%, 90%, 95% power, respectively. The original study used 8 participants (69% power).

Planned Sample

We plan for a sample size of 15 participants (95% power). Participants will mostly be recruited from the Cognitive Science study program at the University of Osnabrück. As in the original study, this is a convenience sample of undergraduate students. No pre-selection rule is used.

In case a participant decides to stop the experiment early, data for this participant will be discarded entirely, and a new participant will be recruited. All participants will be debriefed after the experiment in order to document their impression of the task, whether they noticed regularities, if they pursued any strategy, and how they judged their performance.

Materials

"Stimuli were presented with custom software written with the VisionShell graphics libraries (Comtois, 2004) on Apple desktop computers. Observers sat approximately 46 cm from the monitor without head restraint. Displays were constructed from a set of 12 nonsense shapes used previously in studies of VSL (Figure 1b; e.g., Fiser & Aslin, 2001). Each shape appeared in one or more of 13 colors (including black; see Figure 1c), and subtended roughly 3.3°. Each sequence began with the object located in the center of the display, where it immediately began moving to either the left or right at 11.48°/s. When the center of the object was 12.76° from the display border, the object immediately reversed direction and moved toward the same point on the opposite side of the screen. The object then continued back toward the center of the display, at which time the entire movement could be repeated. A stationary gray occluder, subtending 9.51° x 5.68°, was always present in the center of the display. The object was (at least partially) visible for 1 s between each moment of complete occlusion (see Figure 1a)." (p. 400f)

We adhered to this description with the following exceptions: our stimulus size was 4.3° instead of 3.3°, and point of return was 22.38° from display border instead of 12.76°. These differences are due to different screen resolution and different screen dimensions.

The authors provided the source code to their experiments. However, as the code made use of the proprietary VisionShell graphics libraries on an old Macintosh operating system that is no longer available, we could not run their code for our replication. We therefore re-implemented the experiment using the Psychophysics Toolbox (PTB-3) with Matlab, closely following the original source code.

Procedure

The "Procedure" section on p. 401, and the "Method" section on p. 404, which was specific to experiment 4b, were followed precisely. Instructions were given in German.

As we had no previous experience with VSL and Experiment 4b is considerably more complicated than a standard VSL experiment we first conducted a pilot study with six participants using the design from Experiment 1. The purpose of this pilot study was to verify that our implementation, choice of parameters, and way of instructing participants is suitable to produce the basic VSL effect. The pilot study replicated the original results of experiment 1 with a slightly smaller effect (76% recall rate as opposed to 80% reported by Turk-Browne et al.).

Analysis Plan

We plan to reproduce the original analysis precisely. That means, we will restrict data analysis to those test trials that contained a real triplet instead of a pseudo-triplet (p. 404), and we will conduct the same two-sided t-test as the original authors.

Differences from Original Study

(1) Stimulus size slightly larger (see Materials section)
(2) Different software used for implementation
(3) Different locale (German students instead of U.S.; German instructions instead of English)

These differences should not affect the replicability of the original result.

(Post Data Collection) Methods Addendum

Actual Sample

The sample consisted of 15 people primarily recruited from the Cognitive Science study program. Informed consent was obtained before the start of the experiment. No participant dropped out during the experiment, and no data were excluded.

Differences from pre-data collection methods plan

none

Results

Data preparation

Data analysis in the original study was restricted to those trials where a real triplet was tested against a foil (16 out of 32 trials). Therefore, those trials were extracted from the overall dataset, and the remaining trials were discarded.

Confirmatory analysis

The main result of the original experiment was the learning effect in the trials containing matched-color triplets, as shown by the above-chance recall rate of 72%. This result was significant in a two-sided t-test, $t(7) = 2.892$, $p = .023$, $d = 1.023$. Conducting this analysis on the replication dataset yields very similar results with a recall rate of 70%, 95% confidence interval [58%, 82%], $t(14) = 3.708$, $p = .002$, $d = .957$. A fixed-effects meta-analysis produces a combined estimate of 71%, 95% CI [62%, 79%], $p < .0001$. We also tested whether there is a difference between the original result and the replication result and found no difference, $t(21) = 0.1$. Hence, we clearly replicated the result of the original study.

Exploratory analyses

One possible concern is to what extent the learning process in experiment 4b might be explicit instead of implicit. The original authors report that out of 72 participants in all five experiments, only 2 became fully aware of the triplet structure during the learning phase (p. 401). In contrast, our debriefing showed that more people reported awareness of either both or one of the two matched-color triplets during familiarization. We therefore conducted follow-up analyses excluding those participants that reported some form of regularity, although these analyses go beyond the scope of a direct replication attempt. The table below shows the recall rate of each participant of the replication, and also if they reported awareness of triplets or other regularities.

Participant no.	Recall rate (%)	Noticed triplets?
1	81	1 triplet
2	63	no
3	94	both triplets
4	81	1 triplet
5	100	both triplets
6	100	reported regularities, but not in the form of triplets
7	38	no
8	56	no
9	50	no
10	50	1 triplet
11	69	no
12	94	noticed pairs
13	50	no
14	75	no
15	50	no
mean	**70**	–-

Excluding participants 3 and 5, who both became fully aware of the matched-color triplets yields a recall rate of 66%, 95% CI [54%, 77%], $t(12) = 2.98$, $p = .012$, $d = .826$. Also excluding participants 1, 4, and 10, who reported awareness of one of the two triplets, results in a recall rate of 64%, 95% CI [50%, 79%], $t(9) = 2.249$, $p = .051$, $d = .711$. In order to test the effect of explicit awareness on participants' performance, we split the data, and put participants 1, 3, 4, 5, 10 in one group and the others in the other group. A t-test between the two groups shows no significant difference, $t(13) = 1.55$, $p = .146$, $d = .847$.

Discussion

Summary of Replication Attempt

The primary result of the replication is an above-chance recall rate in trials containing matched-color triplets in experiment 4b. This result supports the hypothesis that VSL can occur over different features of an object independently, and constitutes a successful replication of the results of the original study.

Commentary

There were more participants in our replication study who reported some form of awareness of the triplets than in the original study. This difference might be due to unintended differences, e.g. a difference in the instructions due to the translation, or a student population

who is more experienced with psychological experiments. It could also be the case that we were more direct in asking for patterns in the stimuli than the original study. In any case, we clearly replicated the effect and we did not find a significant difference between the participants who reported awareness and those that did not. However, as we did have considerably more participants who reported awareness, there is a possibility that in our data there is also explicit learning which would make a difference for the interpretation of the effect.